

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12401384
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Kuboshima; Ryo

Reception device for receiving signals

Abstract

A first mixer and a second mixer subject a received signal to quadrature detection by using a local signal. The first mixer and the second mixer subject the received signal subjected to the quadrature detection to frequency conversion so that, of harmonics of a frequency generated by a reference oscillator oscillation, a harmonic having a frequency closest to a frequency of the received signal subjected to the quadrature detection has a predetermined frequency. A first HPF and a second HPF attenuate the predetermined frequency in the received signal subjected to frequency conversion. The first mixer and the second mixer perform frequency conversion so that the frequency of the received signal is 0 Hz. An adjacent channel attenuation filter attenuates an adjacent channel component in the received signal subjected to frequency conversion.

Inventors:	Kuboshima; Ryo (Yokohama, JP)
Applicant:	JVCKENWOOD Corporation (Yokohama, JP)
Family ID:	1000008780866
Assignee:	JVCKENWOOD Corporation (Yokohama, JP)
Appl. No.:	17/817663
Filed:	August 05, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230038667 A1	Feb. 09, 2023

Foreign Application Priority Data

JP	2021-130293	Aug. 06, 2021
----	-------------	---------------

Publication Classification

Int. Cl.: H04B1/26 (20060101)

U.S. Cl.:

CPC H04B1/26 (20130101);

Field of Classification Search

CPC: H03F (1/3211); H03F (3/45475); H03F (2200/129); H03F (2200/294); H03F (2200/451); H03F (2203/45528); H04B (1/16); H04B (1/26); H03M (3/3287); H03M (3/43); H03M (3/454); H04L (25/02); H04L (27/10); H04L (27/32); H04L (27/36)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5108334	12/1991	Eschenbach	331/22	H04B 1/26
6125135	12/1999	Woo	375/150	G01S 19/32
6404293	12/2001	Darabi	331/37	H03B 21/01
2001/0007151	12/2000	Vorenkamp	348/731	H10D 89/60
2002/0057752	12/2001	Denno	375/346	H03D 3/009
2002/0142748	12/2001	Earls	455/314	H03D 7/161
2004/0057534	12/2003	Masenten	375/316	H03M 3/3287
2004/0087296	12/2003	Park	455/283	H04B 1/109
2004/0213358	12/2003	Patel	375/E7.002	H04N 5/4446
2005/0195336	12/2004	Waight	348/E5.113	H03D 3/007
2009/0258629	12/2008	Ritchey	455/313	H03D 7/161
2010/0110307	12/2009	Leme	348/731	H04N 5/455
2010/0285761	12/2009	Morris	455/113	H04B 1/26
2011/0065406	12/2010	Feng	455/313	H04B 1/1036
2013/0231055	12/2012	Leroux	455/41.2	H04W 4/10
2015/0256374	12/2014	Ohba	375/344	H04B 1/1027
2017/0279950	12/2016	Auranen	N/A	H04M 1/72412
2018/0175905	12/2017	Nabki	N/A	H04L 25/00
2018/0348341	12/2017	Phelan	N/A	G01S 7/023
2019/0200125	12/2018	Hilding	N/A	H04R 5/033
2022/0345090	12/2021	Banerjee	N/A	H03F 3/45475
2023/0100595	12/2022	Hur	455/552.1	H04B 1/006
2024/0007137	12/2023	Nicholls	N/A	H03D 7/161

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2011014961	12/2010	JP	N/A

Primary Examiner: Hu; Jinsong

Assistant Examiner: Hu; Rui M

Attorney, Agent or Firm: Shih IP Law Group, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2021-130293, filed on Aug. 6, 2021, the entire contents of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field

(2) The present invention relates to a reception technology and, more particularly, to a reception device that receives signals.

2. DESCRIPTION OF THE RELATED ART

(3) Generally, a wireless communication device includes a reception circuit that receives signal waves transmitted from outside and converts a received radio frequency (RF) signal into an intermediate frequency (IF) signal by frequency conversion for demodulation and an oscillation circuit that outputs a local signal for performing frequency conversion. The oscillation circuit includes a reference oscillator and performs phase locked loop (PLL) control by using the frequency generated by the reference oscillator by oscillation as a reference frequency. The reference oscillator outputs a reference signal at the reference frequency. When the reference oscillator generates a signal at the reference frequency, it normally generates harmonics thereof as well. Radiation of harmonics to the reception circuit and interference of harmonics with the RF signal may cause suppression of the reception sensitivity. Patent literature 1 describes reducing suppression of the reception sensitivity caused by interference of harmonics, by changing the reference frequency. [Patent literature 1] JP2011-14961

(4) In wireless communication devices as described in patent literature 1, however, it is necessary to set whether to change the reference frequency in accordance with the received frequency.

SUMMARY OF THE INVENTION

(5) A reception device according to an embodiment includes: a reference oscillator; an oscillation circuit that performs PLL control by using a frequency generated by the reference oscillator by oscillation as a reference frequency; a quadrature detector that subjects a receiving signal by using a local signal output from the PLL control in the oscillation circuit; a first complex mixer that subjects an in-phase component and an orthogonal component of the received signal subjected to the quadrature detection by the quadrature detector to frequency conversion so that, of harmonics of the frequency generated by the reference oscillator by oscillation, a harmonic having a frequency closest to a frequency of the received signal subjected to the quadrature detection by the quadrature detector has a predetermined frequency; a first attenuation filter that attenuates the predetermined frequency in the in-phase component of the received signal subjected to the frequency conversion by the first complex mixer; a second attenuation filter that attenuates the predetermined frequency in the orthogonal-phase component of the received signal subjected to the frequency conversion by the first complex mixer; a second complex mixer that subjects the in-phase component of the received signal passed through the first attenuation filter and the orthogonal component of the received signal passed through the second attenuation filter to frequency conversion so that the frequency of the received signal is 0 Hz; a first adjacent channel attenuation filter that attenuates an adjacent channel component in the in-phase component of the received signal subjected to the frequency conversion by the second complex mixer; and a second adjacent channel attenuation filter that attenuates an adjacent channel component in the orthogonal phase component of the received signal subjected to the frequency conversion by the second complex mixer.

(6) Optional combinations of the aforementioned constituting elements, and implementations of the

embodiment in the form of methods, apparatuses, systems, recording mediums, and computer programs, etc. may also be practiced as additional modes of the embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments will now be described by way of examples only, with reference to the accompanying drawings which are meant to be exemplary, not limiting and wherein like elements are numbered alike in several Figures in which:
- (2) FIG. 1 shows a configuration of a reception device according to embodiment 1;
- (3) FIG. 2 shows the spectrum of a received signal;
- (4) FIG. 3 shows the spectrum of a spurious component produced by the reference oscillator;
- (5) FIG. 4 shows the spectrum of the received signal and the spurious component produced by the reference oscillator;
- (6) FIG. 5 shows the spectrum of the received signal and the spurious component produced by the reference oscillator that are subjected to frequency conversion;
- (7) FIG. 6 shows a relationship between the received signal and the spurious component produced by the reference oscillator, which are subjected to frequency conversion, and the HPF;
- (8) FIG. 7 shows the spectrum of the received signal subjected to frequency conversion;
- (9) FIG. 8 shows a configuration of the reception device according to embodiment 2;
- (10) FIG. 9 shows the spectrum of the received signal and the spurious component produced by the reference oscillator that are subjected to frequency conversion;
- (11) FIG. 10 shows a relationship between the received signal and the spurious component produced by the reference oscillator, which are subjected to frequency conversion, and the HPF;
- (12) FIG. 11 shows the spectrum of the received signal passed through the HPF; and
- (13) FIG. 12 shows the spectrum of the received signal subjected to frequency conversion.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

- (14) The invention will now be described by reference to the preferred embodiments. This does not intend to limit the scope of the present invention, but to exemplify the invention.

Embodiment 1

(15) The knowledge that provides a basis for the embodiment will be described before describing the embodiment in specific details. The embodiment relates to a reception device that receives radio signals. The reception device generates a local signal by PLL control based on the reference frequency in the reference oscillator and subjects a received signal to quadrature detection by using the local signal. In that process, harmonics generated from the reference oscillator are superimposed on the received signal. As harmonics are superimposed on the received signal, interference between the received signal and the spurious component occurs after the detection. The spurious component having a frequency remote from the received signal is attenuated by an adjacent channel attenuation filter so that the impact of that spurious component on the reception characteristics is small. Meanwhile, the spurious component having a frequency close to the received signal, which may be included in the pass band of the adjacent channel attenuation filter, is not attenuated by the adjacent channel attenuation filter. Therefore, that spurious degrades the reception characteristics. It is therefore desired to suppress degradation of the reception characteristics even in the presence of the spurious component having a frequency close to the received signal, which may be included in the pass band of the adjacent channel attenuation filter. It is also desired to automatically reduce suppression of the reception sensitivity due to interference from spurious components.

- (16) FIG. 1 shows a configuration of a reception device **100**. The reception device **100** has a configuration of a LowIF receiver. The reception device **100** includes a reference oscillator **16**, an

oscillation circuit **18**, an antenna **20**, a first filter **22**, a low noise amplifier (LNA) **24**, a second filter **26**, a first mixer **28a** and a second mixer **28b**, which are generically referred to as mixers **28**, a first variable amplifier **30a** and a second variable amplifier **30b**, which are generically referred to as variable amplifiers **30**, a first AAF **32a** and a second AAF **32b**, which are generically referred to as anti-aliasing filters (AAF) **32**, a first ADC **34a** and a second ADC **34b**, which are generically referred to as analog to digital converters **34**, a first mixer **36a** and a second mixer **36b**, which are generically referred to as mixers **36**, a first oscillator **38**, a first HPF **40a** and a second HPF **40b**, which are generically referred to as high pass filters **40**, a first mixer **42a** and a second mixer **42b**, which are generically referred to as mixers **42**, a second oscillator **44**, a first adjacent channel attenuation filter **46a** and a second adjacent channel attenuation filter **46b**, which are generically referred to as adjacent channel attenuation filters **46**, and a demodulator **48**. The oscillation circuit **18** includes a PLL **50**, a loop filter **52**, a voltage-controlled oscillator (VCO) **54**.

(17) The antenna **20** receives a radio frequency (RF) signal from a transmitting device (not shown). The RF signal represents a received signal. The RF signal is, for example, FM-modulated, but the embodiment is non-limiting as to the nature of the RF signal. The antenna **20** outputs the received RF signal to the first filter **22**. The first filter **22** reduces the noise component included in the RF signal. The first filter **22** outputs the RF signal having the reduced noise component (hereinafter, this will also be referred to as “RF signal”) to the LNA **24**.

(18) The LNA **24** amplifies the RF signal from the first filter **22**. The LNA **24** outputs the amplified RF signal to the second filter **26**. The second filter **26** reduces the noise component included in the amplified RF signal. The second filter **26** outputs the RF signal that has the reduced noise component and has been amplified (hereinafter, this will also be referred to as “RF signal”) to the first mixer **28a** and the second mixer **28b**. FIG. 2 shows the spectrum of a received signal **200**. This represents the RF signal output from the second filter **26** of FIG. 1. As shown in the figure, the received signal **200** is, for example, an unmodulated signal. Reference is made back to FIG. 1.

(19) The reference oscillator **16** outputs a frequency that serves as a reference (hereinafter, referred to as “reference frequency”) to the oscillation circuit **18**. The oscillation circuit **18** performs PLL control based on the reference frequency generated by the reference oscillator **16** by oscillation. PLL control is performed by the PLL **50**, the loop filter **52**, and the VCO **54** so that the local signal is output to the first mixer **28a** and the second mixer **28b**. The phase of the local signal output to the second mixer **28b** is shifted by 90 degrees from the phase of the local signal output to the first mixer **28a**. FIG. 3 shows the spectrum of a spurious component **210** produced by the reference oscillator **16**. This represents harmonics generated from the reference oscillator **16** by oscillation. In this case, it is assumed that the frequencies of a plurality of spurious components **210** from the reference oscillator **16** are not equal to the frequency of the received signal **200** of FIG. 2.

(20) FIG. 4 shows the spectrum of the received signal **200** and the spurious component **210** produced by the reference oscillator **16**. The figure shows a signal in which the spurious component **210** produced by the reference oscillator **16** is superimposed on the received signal **200** and is a signal that is input to the first mixer **28a** and the second mixer **28b** of FIG. 1. Reference is made back to FIG. 1.

(21) The first mixer **28a** generates an in-phase component (hereinafter, “I signal”) of a LowIF (hereinafter, referred to as “first LowIF”) signal by multiplying the RF signal from the second filter **26** with the local signal from the oscillation circuit **18**. The first LowIF is a frequency lower than RF and higher than 0 Hz. The frequency of the local signal from the oscillation circuit **18** is set so that the frequency of the I signal generated by the first mixer **28a** is the first LowIF. The first mixer **28a** outputs the I signal to the first variable amplifier **30a**.

(22) The second mixer **28b** generates the orthogonal component (hereinafter, “Q signal”) of the first LowIF signal by multiplying the RF signal from the second filter **26** with the local signal from the oscillation circuit **18**. The Q signal is a signal orthogonal to the I signal. The second mixer **28b** outputs the Q signal to the second variable amplifier **30b**. The first mixer **28a** and the second mixer

28b correspond to a quadrature detector, and the quadrature detector subjects the RF signal to quadrature detection by using the local signal.

(23) The first variable amplifier **30a** adjusts the level of the I signal. The first variable amplifier **30a** outputs the I signal with the adjusted level (hereinafter, also referred to as “I signal”) to the first AAF **32a**. The second variable amplifier **30b** adjusts the level of the Q signal. The second variable amplifier **30b** outputs the Q signal with the adjusted level (hereinafter, also referred to as “Q signal”) to the second AAF **32b**.

(24) The first AAF **32a** performs band limitation of the I signal. The first AAF **32a** outputs the band-limited I signal (hereinafter, also referred to as “I signal”) to the first ADC **34a**. The second AAF **32b** performs band limitation of the Q signal. The second AAF **32b** outputs the band-limited Q signal (hereinafter, also referred to as “Q signal”) to the second ADC **34b**.

(25) The first ADC **34a** subjects the I signal from the first AAF **32a** to analog-digital conversion. The first ADC **34a** outputs the I signal converted into a digital signal (hereinafter, also referred to as “I signal”) to the first mixer **36a**. The second ADC **34b** subjects the Q signal from the second AAF **32b** to analog-digital conversion. The second ADC **34b** outputs the Q signal converted into a digital signal (hereinafter, also referred to as “Q signal”) to the second mixer **36b**. The first ADC **34a** and the second ADC **34b** can be said to be a sampling unit that samples the signal subjected to quadrature detection according to a predetermined timing schedule.

(26) The first oscillator **38** outputs an oscillation signal (hereinafter, “first oscillation signal”) to the first mixer **36a** and the second mixer **36b**. The first mixer **36a** generates a LowIF (hereinafter, referred to as “second LowIF”) I signal (hereinafter, also referred to as “I signal”) by multiplying the I signal from the first ADC **34a** with the first oscillation signal from the first oscillator **38**. The second LowIF is a frequency lower than the first LowIF and higher than 0 Hz. The frequency of the first oscillation signal is set so that the frequency of the I signal generated in the first mixer **36a** is the second LowIF. The first mixer **36a** outputs the I signal to the first HPF **40a**. The second mixer **36b** generates a second LowIF Q signal (hereinafter, also referred to as “Q signal”) by multiplying the Q signal from the second ADC **34b** with the first oscillation signal from the first oscillator **38**. The second mixer **36b** outputs the Q signal to the second HPF **40b**.

(27) FIG. 5 shows the spectrum of the received signal **200** and the spurious component **210** produced by the reference oscillator **16** that are subjected to frequency conversion. Of the plurality of spurious components **210** shown in FIG. 4, FIG. 5 shows only the spurious component **210** closest to the received signal **200**. Further, the received signal **200** corresponds to the I signal and the Q signal described above and has the second LowIF. The second LowIF of the received signal **200** is set by the frequency conversion by the mixers **36** so that the spurious component **210** closest to the received signal **200** is at 0 Hz. Reference is made back to FIG. 1. The first mixer **36a** and the second mixer **36b** may be referred to as first complex mixers. The first complex mixers subject the in-phase component and the orthogonal component of the received signal **200** to frequency conversion so that, of the harmonics of the frequency generated by the reference oscillator **16** by oscillation, the harmonic of the frequency closest to the frequency of the received signal **200** is at 0 Hz.

(28) The first HPF **40a** is a filter that passes the high-frequency component of the I signal from the first mixer **36a**. The first HPF **40a** outputs the high-frequency component of the I signal (hereinafter, also referred to as “I signal”) to the first mixer **42a**. The first HPF **40a** is also referred to as a first attenuation filter, and the first attenuation filter attenuates low frequencies in the in-phase component of the received signal **200** subjected to frequency conversion by the first complex mixers. The second HPF **40b** is a filter that passes the high-frequency component of the Q signal from the second mixer **36b**. The second HPF **40b** outputs the high-frequency component of the Q signal (hereinafter, also referred to as “Q signal”) to the second mixer **42b**. The second HPF **40b** is also referred to as a second attenuation filter, and the second attenuation filter attenuates the frequencies in the orthogonal component of the received signal **200** subjected to frequency

conversion by the first complex mixers. FIG. 6 shows a relationship between the received signal **200** and the spurious component **210** produced by the reference oscillator **16**, which are subjected to frequency conversion, and the HPF **40**. In addition to what is shown in FIG. 5, FIG. 6 shows frequency characteristics **220** of the HPF **40**. The HPF **40** passes the received signal **200** but reduces the spurious component **210**. Reference is made back to FIG. 1.

(29) The second oscillator **44** outputs the oscillation signal (hereinafter, “second oscillation signal”) to the first mixer **42a** and the second mixer **42b**. The first mixer **42a** generates a baseband I signal (hereinafter, also referred to as “I signal”) by multiplying the I signal from the first HPF **40a** with the second oscillation signal from the second oscillator **44**. The frequency of the second oscillation signal is set so that the frequency of the I signal generated in the first mixer **42a** is the baseband. The first mixer **42a** outputs the I signal to the first adjacent channel attenuation filter **46a**. The second mixer **42b** generates a baseband Q signal (hereinafter, also referred to as “Q signal”) by multiplying the Q signal from the second HPF **40b** with the second oscillation signal from the second oscillator **44**. The second mixer **42b** outputs the Q signal to the second adjacent channel attenuation filter **46b**. The first mixer **42a** and the second mixer **42b** are referred to as second complex mixers. The second complex mixers subject the in-phase component of the received signal **200** passed through the first attenuation filter and the orthogonal component of the received signal **200** passed through the second attenuation filter to frequency conversion so that the frequency of the received signal **200** is 0 Hz.

(30) The first adjacent channel attenuation filter **46a** performs band limitation of the I signal. The band limitation attenuates the adjacent channel component. The first adjacent channel attenuation filter **46a** outputs the band-limited I signal (hereinafter, also referred to as “I signal”) to the demodulator **48**. The second adjacent channel attenuation filter **46b** performs band limitation of the Q signal. The band limitation attenuates the adjacent channel component. The second adjacent channel attenuation filter **46b** outputs the band-limited Q signal (hereinafter, also referred to as “Q signal”) to the demodulator **48**. FIG. 7 shows the spectrum of the received signal **200** subjected to frequency conversion. Of the plurality of spurious components **210**, the spurious component **210** closest to the received signal **200** is attenuated by the HPF **40**, the remainder of the spurious components **210** is attenuated by the adjacent channel attenuation filters **46** so that only the received signal **200** remains as shown. Reference is made back to FIG. 1.

(31) The demodulator **48** subjects the I signal from the first adjacent channel attenuation filter **46a** and the Q signal from the second adjacent channel attenuation filter **46b** to demodulation and outputs a sound signal or data from the demodulation. For the demodulation process, a publicly known technology may be used so that a description thereof is omitted.

(32) When the frequency of the RF signal is “400 MHz” and the pass band width of the adjacent channel attenuation filters **46** is “ ± 5 kHz”, for example, the pass band of the adjacent channel attenuation filters **46** will be “400 MHz ± 5 kHz”. When the frequency of the spurious component **210** closest to the RF signal is “400.002 MHz” in this situation, the spurious component **210** is included in the pass band of the adjacent channel attenuation filters **46** and so is not attenuated by the adjacent channel attenuation filters **46**. Meanwhile, the second LowIF is set to 0 Hz in the reception device **100** according to the embodiment so that the spurious component **210** is attenuated by the HPF **40**.

(33) The features are implemented in hardware such as a CPU, a memory, or other LSIs of an arbitrary computer and in software such as a program loaded into a memory. The figure depicts functional blocks implemented by the cooperation of these elements. Therefore, it will be understood by those skilled in the art that the functional blocks may be implemented in a variety of manners by hardware only, software only, or by a combination of hardware and software.

(34) According to the embodiment, the received signal is subject to frequency conversion so that the spurious component having a frequency closest to the frequency of the received signal is at 0 Hz before the received signal is passed through the HPF. Accordingly, the impact from the spurious

component is reduced. Further, since the impact from the spurious component is reduced, suppression of the reception sensitivity caused by interference of harmonics can be automatically reduced. Further, since the received signal passed through the HPF is turned into a baseband signal by frequency conversion so that the received signal can be demodulated.

Embodiment 2

(35) A description will now be given of embodiment 2. Embodiment 2 relates to a reception device like that of embodiment 1. Embodiment 2 differs from embodiment 1 in respect of the local signal that should be used for quadrature detection. To describe it more specifically, the reception device according to embodiment 2 uses a frequency-modulated local signal, by subjecting the control voltage of the reference oscillator to frequency modulation by using a low frequency. By using such a local signal for quadrature detection, the in-phase component and the orthogonal component of the received signal will also be frequency-modulated. The in-phase component and the orthogonal component of the received signal subjected to quadrature detection are processed by the HPF to reduce the spurious component that interferes with the received signal, but the component of the received signal remains. This reduces suppression of the reception sensitivity due to the spurious component and also reduces degradation due to reception distortion. Hereinafter, a difference from the foregoing will mainly be described.

(36) FIG. 8 shows a configuration of the reception device **100**. In the reception device **100**, a modulation signal generator **10** and a third filter **12** are added to the reception device **100** of FIG. 1.

(37) The modulation signal generator **10** outputs a modulation signal for causing the reference oscillator **16** to perform modulation to the reference oscillator **16** via the third filter **12**. The modulation signal from the third filter **12** is input to the controlled voltage terminal of the reference oscillator **16**, and the frequency generated by oscillation in the reference oscillator **16** is frequency-modulated by the modulation signal. When the frequency modulation is performed at a modulation rate defined by a frequency within the band of the loop filter **52**, the local signal output from the oscillation circuit **18** will also be frequency-modulated.

(38) The first mixer **28a** and the second mixer **28b** subject the received signal and the spurious component produced by the reference oscillator to quadrature detection by using the frequency-modulated local signal. As in embodiment 1, the result of quadrature detection is shown as the I signal and the Q signal at the first Low IF. Unlike the case of embodiment 1, the I signal and the Q signal of embodiment 2 are frequency-modulated. The processes in the variable amplifier **30** and the subsequent stages are similar to those of embodiment 1.

(39) FIG. 9 shows the spectrum of the received signal **200** and the spurious component **210** produced by the reference oscillator **16** that are subjected to frequency conversion. The figure shows a signal output from the mixers **36**. Like FIG. 5, FIG. 9 shows only the spurious component **210** closest to the received signal **200**. Further, the received signal **200** corresponds to the I signal and the Q signal described above. However, the received signal **200** is frequency-modulated by using the frequency-modulated local signal.

(40) FIG. 10 shows a relationship between the received signal **200** and the spurious component **210** produced by the reference oscillator **16**, which are subjected to frequency conversion, and the HPF **40**. In addition to what is shown in FIG. 9, FIG. 6 shows the frequency characteristics **220** of the HPF **40**. FIG. 11 shows the spectrum of the received signal **200** passed through the HPF **40**. The first HPF **40a** and the second HPF **40b** eliminate the spurious component **210** that interferes with the received signal **200**. Further, a portion of the received signal **200** is eliminated, but there are also portions that remain. In other words, the portion of the received signal **200** eliminated by the HPF **40** is reduced, regardless of whether the received signal **200** is an unmodulated wave or a modulated wave. This reduces degradation due to reception distortion.

(41) FIG. 12 shows the spectrum of the received signal **200** subjected to frequency conversion. The figure shows a signal passed through the adjacent channel attenuation filters **46**. The mixers **42** convert the received signal **200** into a baseband signal.

(42) According to this embodiment, the reference frequency is frequency-modulated so that the received signal can be frequency-modulated by using the local signal. Further, since the received signal is frequency-modulated by using the local signal, components eliminated by processing the I signal and the Q signal with the HPF can be reduced. Further, since components eliminated by processing the I signal and the Q signal with the HPF are reduced, degradation due to reception distortion can be reduced. Further, since degradation due to reception distortion is reduced, 2 tone/5 tone decoding, etc. can be performed. Further, since the modulation signal to cause the reference oscillator to perform modulation is output to the reference oscillator, modulation in the reference oscillator can be performed.

(43) Described above is an explanation based on an exemplary embodiment. The embodiment is intended to be illustrative only and it will be obvious to those skilled in the art that various modifications to constituting elements and processes could be developed and that such modifications are also within the scope of the present invention.

(44) In the reception device **100** according to embodiments 1, 2, the second LowIF is set by the frequency conversion by the mixers **36** so that the spurious component **210** closest to the received signal **200** is at 0 Hz. However, the embodiment is non-limiting as to the setting, and the second LowIF may be set so that the spurious component **210** closest to the received signal **200** is not at 0 Hz. In that process, an attenuation filter that attenuates the frequency of the spurious component **210** may be used in place of the HPF **40**. According to this variation, the flexibility in configuration can be improved.

Claims

1. A reception device comprising: a reference oscillator; an oscillation circuit that performs phase locked loop (PLL) control by using a frequency generated by the reference oscillator by oscillation as a reference frequency; a quadrature detector that subjects a receiving signal by using a local signal output from the PLL control in the oscillation circuit; an analog to digital converter (ADC) that subjects an in-phase component and an orthogonal component of the received signal subjected to the quadrature detection by the quadrature detector to analog-digital conversion; a first complex mixer that subjects an output signal of the ADC to frequency conversion so that, of harmonics of the frequency generated by the reference oscillator by oscillation, a harmonic having a frequency closest to a frequency of the received signal subjected to the quadrature detection by the quadrature detector has a predetermined frequency; a first high pass filter that attenuates the predetermined frequency in the in-phase component of the received signal subjected to the frequency conversion by the first complex mixer and passes a first signal component of an output signal of the first complex mixer; a second high pass filter that attenuates the predetermined frequency in the orthogonal-phase component of the received signal subjected to the frequency conversion by the first complex mixer and passes a second signal component of an output signal of the first complex mixer; a second complex mixer that subjects the in-phase component of the received signal passed through the first high pass filter and the orthogonal component of the received signal passed through the second high pass filter to frequency conversion so that the frequency of the received signal is 0 Hz; a first adjacent channel attenuation filter that attenuates an adjacent channel component in the in-phase component of the received signal subjected to the frequency conversion by the second complex mixer; and a second adjacent channel attenuation filter that attenuates an adjacent channel component in the orthogonal phase component of the received signal subjected to the frequency conversion by the second complex mixer; wherein the first complex mixer subjects the in-phase component and the orthogonal component of the received signal subjected to the quadrature detection by the quadrature detector to the frequency conversion so that, of harmonics of the frequency generated by the reference oscillator by oscillation, a harmonic having a frequency closest to the frequency of the received signal subjected to the quadrature detection by the

quadrature detector is at 0 Hz; the first high pass filter passes a high-frequency component in the in-phase component of the received signal subjected to the frequency conversion by the first complex mixer, and the second high pass filter passes a high-frequency component in the orthogonal component of the received signal subjected to the frequency conversion by the first complex mixer.

2. The reception device according to claim 1, wherein the frequency generated by the reference oscillator by oscillation is modulated within a band of a loop filter of the oscillation circuit.
